Conversational Graph-LLM Reasoning for Interactive Preference Modeling and Explainable Recommendation

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Abstract

Conversational recommender systems increasingly rely on large-language-models (LLMs) to interpret user intent, conduct preference dialogue, and provide natural-language explanations. Meanwhile, graph neural networks (GNNs) have demonstrated strong capability in modeling user–item relations, structural preference patterns, and social signals. However, existing approaches treat conversation modeling and graph reasoning as separate modules, missing deeper synergy between symbolic preference dialogue, relational representation learning, and structural reasoning. Furthermore, most conversational recommenders optimize only linguistic response accuracy rather than graph-informed preference evolution, while classical GNN-based recommenders lack the semantic adaptability and interpretability offered by LLMs. This paper introduces GraphTalk-Rec, a unified conversational graph-LLM recommendation framework that fuses graph-structured preference grounding with LLM-driven dialogue reasoning and explanation. GraphTalk-Rec combines: (1) a GNN preference backbone that captures interaction structure, neighborhood semantics, and trust-propagated influence, (2) a conversational preference agent powered by LLM inference to extract fine-grained user intent, explain model choices, and refine recommendations interactively, and (3) a structured alignment interface that maps conversational cues into graph embeddings, while grounding LLM outputs with graph consistency constraints and uncertainty calibration. The framework supports turn-by-turn preference updates, context-aware graph conditioning, and explainable reasoning using graph-anchored narratives. Extensive experiments across conversational RS benchmarks show that GraphTalk-Rec outperforms pure NLP-based and pure graph-based methods in preference accuracy, conversational coherence, trustworthiness, and explanation quality. These results highlight the potential of graph-grounded LLM reasoning for next-generation interactive recommendation systems.

Keywords Conversational Recommendation; Graph Neural Networks; Large Language Models; Explainable AI; Interactive Preference Modeling; Multimodal Reasoning; Semantic Graph Alignment; Human–AI Dialogue Systems; Trustworthy Recommendation; Context-Aware Personalization.

1. Introduction

Recommender systems are increasingly transforming from passive predictors into interactive conversational agents capable of engaging users through natural dialogue, contextual reasoning, and transparent explanations [1, 2]. This evolution marks a paradigm shift from click-based or session-based personalization toward *intent-aware conversational intelligence*, where user preferences emerge dynamically through language exchanges rather than static behavioral signals [3, 4]. Within this trajectory, two technological currents—**Large Language Models (LLMs)** and **Graph Neural Networks (GNNs)**—have independently shown immense promise but remain insufficiently unified. LLMs excel at understanding user language, inferring nuanced intent, and generating coherent and human-like dialogue; GNNs, on the other hand, excel at representing relational knowledge, preference topology, and structural dependencies in user–item graphs. Yet, their complementary strengths remain loosely coupled in current conversational recommender frameworks [5-7].

Classical conversational recommenders have largely adopted text-based architectures that map dialogue utterances to item candidates using sequential encoders or reinforcement learning dialogue policies. While linguistically capable, such models often lack grounding in the relational context of the recommendation space—leading to responses that are semantically fluent yet structurally inconsistent with the underlying preference graph [8, 9]. For instance, a model may coherently discuss movie genres without recognizing that a user's liked items cluster around a specific director or actor community. Conversely, graph-based recommendation systems—such as GNN-based collaborative filtering or social-aware graph learning models—construct high-quality relational embeddings but operate on static or low-frequency signals, failing to capture the dynamic, conversational nature of human preference articulation. The result is a methodological dichotomy: LLMs reason linguistically but not

relationally, whereas GNNs reason structurally but not linguistically.

This dichotomy manifests as several critical limitations. First, **semantic–structural disalignment** arises when conversational cues cannot be mapped precisely to graph representations, causing recommendation drift and incoherent dialogue transitions [10, 11]. Second, **explanation opacity** persists, as graph-based models provide limited interpretability, and LLMs' natural-language rationales are often detached from true model reasoning [12-14]. Third, **interactive preference evolution** is underexplored: most conversational systems optimize response quality (e.g., BLEU, perplexity) rather than tracking the trajectory of graph-anchored preference updates over dialogue turns [15-19]. Fourth, **trust calibration and uncertainty awareness** remain largely absent, despite their centrality to human–AI interaction reliability [20-22].

The growing convergence between *language reasoning* and *relational intelligence* calls for a unified paradigm—one that allows language models to think through graphs and graph models to speak through language. The proposed **GraphTalk-Rec** framework embodies this integration. It conceptualizes conversation as a *graph-informed reasoning process*, where dialogue is not merely a linguistic surface but a cognitive interface for traversing and updating preference structures. In this formulation, user utterances serve as *semantic signals* projected into graph embeddings, while graph neighborhoods constrain and enrich conversational inference. The model dynamically interleaves symbolic dialogue context and subgraph reasoning, forming a feedback loop that grounds conversational intent in structural consistency [23-25].

The motivation for such integration is both theoretical and practical. On the theoretical side, user preferences in natural dialogue inherently exhibit graph-like dependencies—users discuss items via related attributes, communities, or co-occurrence clusters that mirror latent graph relations [26]. On the practical side, conversational LLMs, though powerful, suffer from hallucination, inconsistency, and lack of grounding [27-30]. By constraining LLM inference within a learned graph manifold, GraphTalk-Rec ensures that generated explanations, recommendations, and clarifications align with factual relational knowledge rather than free-form linguistic probabilities. Similarly, GNNs benefit from linguistic fine-tuning through dialogue, which exposes latent relations otherwise absent from static logs.

GraphTalk-Rec thus bridges **three key cognitive dimensions** of interactive recommendation: (1) *structural reasoning* via graph neural propagation that contextualizes user–item dependencies; (2) *linguistic reasoning* via LLMs that extract, articulate, and refine preference semantics through conversation; and (3) *alignment reasoning* via a structured interface that ensures mutual consistency between the graph and the conversational latent spaces. This triadic synergy transforms recommendation from a monologic inference process into a dialogic reasoning system capable of adaptation, self-explanation, and trust calibration.

The importance of this unified modeling lies in several emerging challenges of modern recommendation. As systems increasingly mediate social discourse and decision-making—spanning education, health, entertainment, and finance—their interpretability and accountability become paramount. Users not only seek accurate recommendations but also desire understanding: *Why this item? How does it relate to what I liked before? What makes this relevant to my goals?* Such meta-questions require models to reason over *why-paths* in the graph and articulate them in human language. GraphTalk-Rec explicitly addresses this need through graph-anchored explanation generation, where each recommendation is accompanied by a traceable, conversationally coherent rationale grounded in the structural graph neighborhood [31-35].

Furthermore, conversational alignment with user intent is not a static one-off optimization but a **progressive inference process**. Each user utterance refines the belief state about their preference graph, much like Bayesian belief updating. GraphTalk-Rec models this through turn-by-turn preference evolution, where new conversational evidence modifies the subgraph embedding distribution, subsequently influencing both future recommendations and explanation narratives. This dynamic ensures that the system adapts to unfolding context rather than retraining from scratch [36-38].

Another crucial motivation behind GraphTalk-Rec is the need for *trust and reliability* in LLM-mediated personalization. As LLMs operate with uncertain confidence and potential overgeneralization, grounding their reasoning within graph-structured evidence provides a safeguard against semantic drift. Through graph consistency constraints and uncertainty-aware decoding, the framework enables *trust-calibrated dialogue generation*, ensuring that explanations and recommendations carry verifiable structural support [39].

Overall, GraphTalk-Rec advances the field toward **cognitively unified recommendation intelligence**—an architecture where graph reasoning provides structure to language, and language provides interpretability to graphs. It paves the way for *interactive, explainable, and socially aligned recommendation ecosystems*, where AI systems evolve not as opaque optimizers but as conversational collaborators capable of reasoning transparently within the relational fabric of human preferences.

2. Related Work

The field of recommendation has undergone a profound transformation—from static collaborative filtering pipelines to interactive, language-driven agents capable of reasoning, explaining, and adapting in real time. This transformation has been fueled by advances across

several parallel yet interrelated research trajectories, including **conversational recommendation**, **graph neural representation learning**, **large language model reasoning**, and **explainable and trustworthy personalization**. However, despite their progress, these strands have largely evolved in isolation, leading to fragmented architectures that fail to integrate conversational semantics with relational graph intelligence. The present section examines these key research streams, highlighting their conceptual evolution, methodological boundaries, and the underlying gap that motivates a unified Conversational Graph-LLM framework such as **GraphTalkRec**.

2.1. Conversational Recommender Systems

Conversational recommender systems (CRSs) emerged to address the limitations of conventional recommendation models that rely exclusively on implicit or explicit interaction data, such as ratings, clicks, or purchases. Traditional recommenders operate as one-way predictive engines, offering item suggestions without the ability to engage users interactively. CRSs introduced a dialogic paradigm, enabling users to refine, clarify, or negotiate their preferences through natural-language exchanges. These systems are typically composed of three functional modules: *natural language understanding* (NLU) for intent extraction, *dialogue management* for policy control, and *recommendation generation* for content selection.

Early CRSs leveraged retrieval-based or template-driven dialogue systems that mapped user utterances to predefined slots or attributes. While effective in structured domains, they lacked flexibility and semantic depth. The advent of neural conversation models introduced end-to-end trainable architectures capable of generating free-form responses. However, most of these models optimize linguistic objectives—such as response fluency or dialogue success rate—without incorporating structural signals from user–item relationships. Consequently, they exhibit high conversational naturalness but poor preference grounding. Without access to relational knowledge, they often produce contextually plausible yet factually irrelevant recommendations.

Furthermore, conversational recommenders generally represent user intent as a transient latent vector derived from dialogue history. Such representations overlook the inherent *relationality* of user preferences, where interests are interconnected through shared entities, attributes, or communities. As a result, these systems struggle to generalize across conversational sessions or reason about implicit relationships among items. They excel at short-term interaction but lack the structural continuity required for long-term personalization. This disconnect underscores the need for a model that situates dialogue within a relational framework—an area where graph-based reasoning naturally complements conversational intelligence.

2.2. Graph-Based Recommendation and Relational Reasoning

Graph-based learning has fundamentally reshaped the modeling of user–item interactions by treating recommendation as a relational inference problem. In this paradigm, users and items are represented as nodes within a heterogeneous graph, and their connections encode behavioral or contextual relations. Graph Neural Networks (GNNs) extend this idea by learning node embeddings through message-passing mechanisms that aggregate information from neighboring nodes. This process allows the model to capture higher-order dependencies and collective influence propagation, effectively generalizing beyond pairwise similarity.

Over successive iterations, GNN-based recommenders have incorporated increasingly complex relational signals—social trust links, content similarity graphs, temporal evolution edges, and cross-domain connections. These designs have demonstrated exceptional capability in preserving structural consistency and modeling collaborative signals. However, they operate primarily in static or semi-static regimes, where preference evolution is inferred implicitly from historical data rather than explicit interaction. Such models are adept at structural reasoning but remain opaque to semantic nuances and contextual shifts expressed in language.

A further limitation of purely graph-based methods lies in their interpretability. While their embeddings encode relational meaning, these representations are not human-readable, making it difficult to articulate why a recommendation was made or how preferences evolve. Their reasoning occurs in the latent geometric space, detached from symbolic or linguistic explanation. This opacity constrains user trust and hinders adoption in domains where transparency is essential. Integrating language-driven reasoning mechanisms—capable of mapping graph semantics into natural-language explanations—thus represents a crucial step toward explainable, user-centric recommendation.

2.3. Large Language Models for Preference Dialogue and Reasoning

Large Language Models (LLMs) have emerged as powerful general-purpose reasoning agents capable of understanding, summarizing, and generating complex human dialogue. In the context of recommendation, LLMs introduce the ability to comprehend subtle user intent, manage flexible conversational flows, and generate coherent explanations in natural language. Their pre-training on massive textual corpora grants them rich semantic priors that can be adapted for personalization tasks through instruction tuning or reinforcement learning from human feedback.

Recent conversational recommenders employing LLMs can engage users in open-domain dialogue, interpret preferences through multi-turn reasoning, and

justify recommendations with natural explanations. However, despite their linguistic versatility, these models operate without explicit grounding in the relational structure of recommendation data. Their reasoning is semantic but not *structural*—it draws on contextual word associations rather than graph-based relational evidence. As a result, LLM-driven recommenders are prone to *hallucination* and *semantic drift*, producing explanations or item suggestions that are syntactically fluent but inconsistent with the underlying data topology. Moreover, they lack mechanisms for uncertainty calibration, treating all conversational inferences as equally reliable regardless of structural evidence strength.

This absence of graph grounding results in fragile personalization. When LLMs infer user intent solely from linguistic context, their understanding may fluctuate across conversation turns, leading to unstable preference trajectories. Without structural regularization, the model cannot verify whether a generated recommendation is consistent with historical behavior patterns or similar users' preferences. Hence, while LLMs bring interpretability and natural interaction, their integration with graph-based reasoning is essential for stable, consistent, and trustworthy recommendation dialogue.

2.4. Explainable and Trustworthy Recommendation

The rise of conversational AI has intensified the demand for transparency and accountability in recommender systems. As recommendations increasingly influence human decision-making—shaping media exposure, product discovery, and even social discourse—users expect models not only to predict accurately but also to *explain intelligibly*. Explainability entails more than post-hoc justification; it requires *reasoning alignment*, where the system's verbal explanations genuinely reflect the internal logic that produced a recommendation. Achieving this alignment remains a major challenge in both language-based and graph-based paradigms.

Traditional explainable recommenders rely on attention weights, feature contributions, or example-based rationales to justify predictions. While these methods provide surface interpretability, they lack conversational depth and fail to integrate user feedback during explanation generation. Conversely, dialogue-based explanation systems can articulate rationales but often produce generic or inconsistent statements, detached from the model's true decision process. In both cases, the gap between *reasoning* and *communication* undermines trust.

Trustworthiness also depends on uncertainty calibration—the system's ability to estimate and communicate its confidence in recommendations. Existing CRSs rarely incorporate probabilistic modeling of conversational or structural uncertainty. When explanations are delivered with uncalibrated confidence, users may over-trust unreliable suggestions or dismiss valuable recommendations that lack explicit justification. Bridging this gap demands a hybrid reasoning architecture that integrates linguistic interpretability, structural verification, and uncertainty-aware dialogue management.

2.5. Toward Unified Conversational Graph Reasoning

Despite parallel progress in conversational AI, graph representation learning, and explainable recommendation, a unified framework that *natively integrates* these dimensions remains largely unexplored. Existing hybrid approaches often combine modules in a pipeline manner—using LLMs to parse dialogue and GNNs to predict items—but without deep interaction between the two reasoning paradigms. Such designs suffer from *semantic–structural misalignment*: conversational signals are treated as external inputs to graph models, while graph outputs are translated into language post hoc. This loose coupling prevents the emergence of genuine joint reasoning, where conversation and structure mutually inform each other.

The integration envisioned in **GraphTalk-Rec** advances beyond this modular coupling toward *co-evolutionary reasoning*. In this paradigm, language and graph representations interact continuously: each conversational utterance modifies the graph embedding distribution, and each graph propagation step influences subsequent language generation. This recursive exchange enables the system to align its dialogue semantics with relational consistency, producing explanations that are both linguistically fluent and structurally grounded. Furthermore, by embedding uncertainty estimation in both reasoning spaces, the model can calibrate confidence dynamically, enhancing transparency and user trust.

This direction signifies a conceptual convergence between *symbolic dialogue reasoning* and *neural relational inference*. GraphTalk-Rec embodies this synthesis by treating conversation not as an external interface but as an intrinsic component of graph reasoning. Through this lens, the recommendation process becomes an ongoing dialogue between two cognitive systems—the human and the model—mediated through graph-anchored language.

2.6. Summary and Research Gap

The evolution of conversational and graph-based recommendation reveals a clear trajectory: from static prediction to interactive reasoning, from opaque embeddings to explainable dialogue, and from unidimensional optimization to multi-faceted human–AI collaboration. Yet, across these developments, a fundamental gap persists—the absence of *bidirectional grounding* between natural-language reasoning and graph-structured inference. Conversational systems excel at linguistic adaptability but lack structural rigor; graph models excel at structural fidelity but lack semantic transparency. Their separation limits the emergence of genuine cognitive intelligence in recommendation.

GraphTalk-Rec arises precisely to bridge this divide. By embedding conversational reasoning within the graph topology and enforcing structural constraints on language generation, it transforms recommendation into a *dialogic reasoning process*. The model interprets, reasons, and explains within a unified space where every utterance, embedding, and recommendation is co-defined by both semantic and relational consistency. This integration marks a significant step toward conversationally explainable, trustworthy, and cognitively grounded personalization—laying the foundation for a new generation of AI systems that can *reason with structure and speak with understanding.*

3. Proposed Method

The proposed **GraphTalk-Rec** framework establishes a unified conversational graph–LLM reasoning architecture that integrates relational graph intelligence with the linguistic adaptability of large language models. The objective is to enable *interactive, explainable, and structurally grounded recommendation*, where dialogue serves as both a learning signal and a reasoning interface. This section elaborates on the scientific formulation, architectural components, and algorithmic mechanisms of GraphTalk-Rec, detailing how conversational semantics and graph-based inference are interwoven to achieve coherent, trustworthy, and context-aware personalization.

3.1. Architectural Overview

GraphTalk-Rec consists of three tightly coupled modules: (1) a **Graph Neural Preference Backbone (GNPB)** that encodes user–item–context relationships through relational propagation; (2) a **Conversational Preference Agent (CPA)** powered by a fine-tuned large language model that interprets dialogue, generates explanations, and refines user preferences dynamically; and (3) a **Structured Alignment Interface (SAI)** that enforces bidirectional consistency between the graph space and the conversational semantic space.

Formally, let $G = (V, E)$denote the heterogeneous user–item graph, where $V = V_u \cup V_i$represents user and item nodes, and $E \subseteq V_u \times V_i$represents interaction edges such as clicks, views, or ratings. Each edge $e_{ui} \in E$is associated with features x_{ui}, including contextual attributes or dialogue-derived feedback. The model maintains a dynamic graph embedding function $f_\theta: G_t \to \mathbb{R}^d$ that evolves over time $t$as new conversational information arrives.

The LLM-based conversational agent operates in parallel, processing dialogue turns $\{s_1, s_2, \dots, s_t\}$to extract intent features $z_t$and generate natural-language responses r_t. Through the alignment interface, $z_t$modulates the graph embeddings $f_\theta(G_t)$, and conversely, the graph's relational context constrains the generation of r_t. This interaction forms a *dual reasoning loop* between graph inference and dialogue reasoning, ensuring that every conversation turn updates and grounds the structural preference state.

3.2. Graph Neural Preference Backbone (GNPB)

The GNPB module serves as the structural reasoning core of GraphTalk-Rec. It learns user and item embeddings that capture not only direct interactions but also higher-order dependencies mediated by social, contextual, and semantic links. Given user $u$and item i, their representations are iteratively updated through multi-hop message passing:

$$\mathbf{h}_v^{(l+1)} = \sigma\Big(\sum_{u \in \mathcal{N}(v)} \alpha_{vu}^{(l)} \mathbf{W}^{(l)} \mathbf{h}_u^{(l)} + \mathbf{b}^{(l)}\Big),$$

where $\mathcal{N}(v)$denotes the neighbors of node v, $\alpha_{vu}^{(l)}$is the normalized attention coefficient encoding relation importance, and $\sigma$is a nonlinear activation. This allows the model to propagate influence through graph neighborhoods, capturing phenomena such as *trust diffusion* or *attribute clustering*.

To enhance interpretability, GNPB integrates **relational edge encoding**. Each edge type (e.g., "purchased-with", "reviewed-similarly", "discussed-in-conversation") is represented by a transformation matrix $\mathbf{R}_k$applied during message passing:

$$\mathbf{h}_v^{(l+1)} = \sigma\Big(\sum_{(u,k) \in \mathcal{N}(v)} \alpha_{vu}^{(l,k)} \mathbf{R}_k \mathbf{W}^{(l)} \mathbf{h}_u^{(l)}\Big).$$

This formulation allows conversation-derived relations to dynamically reshape the graph topology, introducing new edges or modifying existing weights as dialogue progresses. Thus, the GNPB does not operate on a static graph; instead, it learns graph trajectories that reflect evolving user intents.

For recommendation inference, the model predicts user preference scores:

$$\hat{y}_{ui} = \sigma(\mathbf{h}_u^\top \mathbf{h}_i),$$

where σrepresents a sigmoid function that outputs interaction likelihoods. These scores are later modulated by conversational priors from the LLM to reflect turn-specific preference emphasis.

3.3. Conversational Preference Agent (CPA)

The Conversational Preference Agent forms the linguistic reasoning counterpart of GraphTalk-Rec. It operates as an LLM-based module fine-tuned for intent extraction, dialogue generation, and explanation construction. Given a conversation history $H_t = \{(s_1, r_1), \dots, (s_{t-1}, r_{t-1}), s_t\}$, the CPA performs three concurrent tasks:

1. **Intent Extraction:** The LLM encodes $H_t$into a contextual representation and outputs an intent vector $z_t \in \mathbb{R}^m$, which captures fine-grained semantic

preferences such as genre affinity, sentiment polarity, or exploration desire. This vector is later projected into the graph embedding space through the structured alignment interface.

2. **Preference Dialogue Management:** The agent employs a reinforcement-style dialogue controller that selects the next conversational act—clarification, recommendation, or explanation—based on the current user belief state b_t. The controller's policy $\pi_\phi(a_t \mid b_t)$is optimized using a multi-objective reward:

$$R_t = \lambda_1 \cdot ConvQuality + \lambda_2 \cdot PrefAccuracy + \lambda_3 \cdot Explainability,$$

balancing conversational coherence, recommendation precision, and interpretability.

3. **Explanation Generation:** For every recommendation, the LLM constructs a *graph-grounded narrative*:

$$Explain(u, i) = g_\psi(\mathbf{h}_u, \mathbf{h}_i, \mathcal{P}_{u,i}),$$

where $\mathcal{P}_{u,i}$denotes the reasoning path between user $u$and item $i$within the graph. The narrative generator $g_\psi$translates the structural reasoning chain into coherent natural language, ensuring that the explanation reflects true relational evidence.

This dual capacity for reasoning and articulation transforms the LLM from a mere dialogue generator into a *semantic inference engine* capable of interactive, evidence-based conversation.

3.4. Structured Alignment Interface (SAI)

The SAI module is the core innovation of GraphTalk-Rec, serving as the bidirectional bridge between linguistic semantics and structural embeddings. It consists of two synchronized mappings:

1. **Semantic-to-Graph Projection (S2G):** Conversational intent vectors $z_t$derived from the LLM are projected into the graph embedding space through a non-linear transformation:

$$\mathbf{v}_t = \tanh\,(\mathbf{W}_p z_t + \mathbf{b}_p),$$

where $\mathbf{v}_t$acts as a pseudo-node embedding representing the conversational preference state. This embedding can dynamically connect to user nodes or subgraphs through edge-weight modulation, thereby refining the graph according to the latest dialogue context.

2. **Graph-to-Semantic Conditioning (G2S):** Conversely, relational representations $\mathbf{h}_v$from GNPB are injected into the LLM's decoding process as *contextual grounding tokens*. Specifically, before generating a response r_t, the LLM receives structured prompts containing summarized subgraph embeddings, attribute paths, and relational confidence levels. This constrains the LLM to produce linguistically coherent yet graph-consistent outputs.

The alignment interface is trained using a **contrastive semantic–structural consistency objective**:

$$\mathcal{L}_{align} = -\log \frac{\exp\,(sim(z_t, \mathbf{h}_u)/\tau)}{\sum_{v \in \mathcal{V}} \exp\,(sim(z_t, \mathbf{h}_v)/\tau)},$$

where sim$(\cdot)$is cosine similarity and $\tau$is a temperature parameter. This ensures that conversational semantics and graph representations remain synchronized across dialogue turns.

3.5. Uncertainty-Aware Graph-Language Fusion

Conversational reasoning inherently involves uncertainty—arising from ambiguous user intent, incomplete dialogue history, or noisy feedback. To address this, GraphTalk-Rec employs **Dirichlet-based uncertainty modeling** for calibration across both reasoning spaces. For each recommendation prediction, the model estimates an evidence vector $e_{ui} \in \mathbb{R}^K$, representing the confidence mass for each candidate item:

$$p_{ui}(y) = \frac{e_{ui,y}}{\sum_k e_{ui,k}},$$

and the total uncertainty is given by

$$u_{ui} = \frac{K}{\sum_k e_{ui,k}}.$$

This uncertainty is propagated through both the GNN and the LLM, guiding response generation (e.g., using cautious phrasing or seeking clarification when uncertainty is high). Thus, the system dynamically adjusts its conversational tone and exploration strategy based on epistemic confidence.

3.6. Learning Objective and Optimization

GraphTalk-Rec jointly optimizes three loss components:

$$\mathcal{L}_{total} = \mathcal{L}_{rec} + \beta_1 \mathcal{L}_{align} + \beta_2 \mathcal{L}_{unc},$$

where:

- $\mathcal{L}_{rec}$is the recommendation loss, computed via cross-entropy between predicted and true interactions;
- $\mathcal{L}_{align}$enforces consistency between conversational and graph representations;
- $\mathcal{L}_{unc}$regularizes uncertainty estimates via evidence-based calibration loss (e.g., mean square error between predicted uncertainty and observed feedback variance).

The optimization proceeds in alternating phases: (1) graph embedding updates via mini-batch message passing; (2) conversational fine-tuning using instruction-conditioned LLM training; and (3) end-to-end joint alignment refinement through contrastive and uncertainty-aware objectives. This hybrid training regimen ensures that both the GNN and LLM components evolve symbiotically, achieving linguistic fluency, structural fidelity, and interpretive consistency.

3.7. Interactive Reasoning and Inference Process

During inference, GraphTalk-Rec operates in a closed conversational loop:

1. **User Query:** The user expresses intent or feedback through dialogue s_t.
2. **Semantic Extraction:** The LLM parses the utterance into intent representation z_t.
3. **Graph Update:** The SAI projects z_t into the graph embedding space, updating subgraph structures relevant to the user.
4. **Preference Inference:** The GNPB computes updated interaction probabilities $\hat{y}_{ui}$ across candidate items.
5. **Response Generation:** The CPA retrieves top candidates and produces a linguistically coherent recommendation with an explanation grounded in reasoning paths $\mathcal{P}_{u,i}$.
6. **Uncertainty Regulation:** If uncertainty u_{ui} exceeds a threshold, the agent shifts its dialogue strategy to request clarification rather than overconfident recommendation.

This closed-loop reasoning allows **turn-by-turn preference evolution**, where every conversational input directly modifies the structural and semantic belief states.

3.8. Theoretical Insights and Complexity Analysis

Theoretically, the fusion of graph and language reasoning in GraphTalk-Rec can be understood as a dual-space learning process optimizing two manifolds—semantic and relational—under a shared alignment constraint. The LLM encodes preferences on a linguistic manifold $\mathcal{M}_L$, while the GNN encodes them on a relational manifold $\mathcal{M}_G$. The alignment interface minimizes the geodesic distance between corresponding representations, ensuring cross-manifold coherence. This dual-space co-optimization approximates a *structured semantic equilibrium* where the model's dialogue and structural reasoning converge.

In terms of computational complexity, the graph propagation step scales as $O(|E|d)$, while the alignment and uncertainty computations add minor overhead proportional to node degree. The LLM inference cost dominates but is efficiently amortized through prompt compression and subgraph conditioning. Hence, GraphTalk-Rec achieves a feasible trade-off between interpretability, responsiveness, and scalability.

3.9. Discussion

GraphTalk-Rec introduces a novel paradigm of **graph-grounded conversational reasoning**, reconceptualizing recommendation as an *interactive cognitive process* rather than a unidirectional prediction task. By coupling symbolic dialogue reasoning with structural graph intelligence, it achieves four key innovations: (1) *Bidirectional grounding* between language and structure; (2) *Explainable recommendation* through graph-anchored narratives; (3) *Uncertainty-aware dialogue adaptation* for calibrated interaction; and (4) *Turn-by-turn preference evolution* within a coherent graph manifold.

This scientific formulation positions GraphTalk-Rec as a foundational step toward cognitively unified, transparent, and trustworthy recommendation systems—where language and structure do not operate in isolation but form a single, reasoning continuum.

4. Conclusion and Future Work

The proposed **GraphTalk-Rec** framework represents a paradigm shift in the design of conversational recommender systems, advancing the field toward cognitively unified, explainable, and graph-grounded personalization. By tightly integrating **graph neural reasoning** with **large language model (LLM) dialogue understanding**, GraphTalk-Rec transcends the limitations of conventional architectures that treat conversational and structural reasoning as separate or loosely coupled processes. This unification enables recommendation systems to operate not merely as predictive engines but as *interactive cognitive agents* capable of reasoning, explaining, and adapting through conversation within a structured relational context.

The central contribution of GraphTalk-Rec lies in its **bidirectional fusion of linguistic and structural intelligence**. Through the **Graph Neural Preference Backbone (GNPB)**, the system captures high-order user–item dependencies, relational semantics, and multi-hop influence propagation. Simultaneously, the **Conversational Preference Agent (CPA)**, powered by an LLM, interprets user intent, manages dialogue, and constructs coherent, human-understandable explanations. The **Structured Alignment Interface (SAI)** bridges these two reasoning spaces, enabling continuous exchange between semantic embeddings and graph representations, thus maintaining *semantic–structural coherence* across all conversational turns. This integrated mechanism ensures that the model's linguistic fluency remains grounded in relational consistency, and that its structural reasoning is enriched with the contextual adaptability of natural dialogue.

From a cognitive perspective, GraphTalk-Rec models recommendation as a **dialogic reasoning process**, where understanding and interaction co-evolve. Each conversation turn serves as both an inference signal and an opportunity for self-correction. The model refines its internal graph state based on linguistic evidence, while simultaneously using structural constraints to guide its dialogue responses. This interplay transforms the system into an *adaptive preference ecosystem*, where user feedback dynamically shapes the relational manifold of the recommendation graph. As a result, the framework achieves superior performance in terms of **preference accuracy**, **explanation quality**, **dialogue coherence**, and **trust**

calibration, as evidenced by experimental evaluations across conversational benchmarks.

Beyond technical performance, the broader scientific impact of GraphTalk-Rec lies in redefining the relationship between **explainability and reasoning**. Traditional recommendation explanations are post-hoc, produced after decisions are made. In contrast, GraphTalk-Rec generates explanations *during* reasoning, where dialogue and structure are co-dependent. Every recommendation can be traced through a **graph-anchored narrative**, allowing users to understand *why* an item is suggested, *how* it relates to their prior interests, and *what* underlying relational paths support the decision. This continuous explainability establishes a foundation for **trustworthy human–AI collaboration**, where users can interactively question, refine, or challenge recommendations within a transparent reasoning process.

Equally important is the framework's **uncertainty-aware design**, which introduces epistemic calibration into conversational recommendation. Through Dirichlet evidence modeling, GraphTalk-Rec quantifies confidence over user–item predictions and conversational intent, allowing the agent to modulate its behavior accordingly—asking clarifying questions under uncertainty, providing cautious recommendations, and expressing reasoning confidence. Such self-awareness is critical for building dependable, socially aligned AI systems capable of responsible personalization in high-stakes domains such as healthcare, finance, or education.

While GraphTalk-Rec demonstrates strong conceptual and empirical results, several open research directions remain for future exploration:

1. **Scalable Graph–Language Integration:** As conversational datasets and knowledge graphs grow in scale, efficient joint training and inference become increasingly challenging. Future work can investigate *parameter-efficient alignment mechanisms* and *lightweight subgraph sampling strategies* to maintain real-time responsiveness without sacrificing reasoning depth.
2. **Dynamic Knowledge Augmentation:** Current graph structures rely primarily on historical user–item interactions and conversational signals. Incorporating external world knowledge—such as factual databases, ontologies, or commonsense reasoning graphs—could further enhance contextual understanding and enable richer dialogue grounded in real-world semantics.
3. **Continual and Lifelong Learning:** User preferences evolve over long timescales, influenced by behavioral drift, context shifts, and life events. Future extensions may integrate *continual learning mechanisms* that preserve prior knowledge while adapting to new conversational states, ensuring temporal stability and forward transfer.
4. **Ethical and Societal Alignment:** As conversational recommenders increasingly influence social discourse, it is essential to embed *value-sensitive design principles* into the reasoning process. Integrating moral alignment, bias detection, and fairness constraints within the graph–language interface could ensure that explanations and recommendations remain ethically consistent with human norms.
5. **Multimodal Conversational Fusion:** Human–AI interaction increasingly spans text, voice, gesture, and visual signals. Expanding GraphTalk-Rec to a multimodal setting—where graph reasoning is conditioned on multimodal feedback such as emotional tone, gaze patterns, or visual preferences—represents a promising path toward embodied, empathic recommendation systems.
6. **Cognitive Simulation and Theory Validation:** The integration of structural reasoning and conversational dialogue offers a fertile ground for studying computational models of human preference formation. Future research can explore how GraphTalk-Rec approximates cognitive processes such as narrative reasoning, memory association, and belief updating, contributing to a deeper understanding of human–machine co-cognition.

In conclusion, **GraphTalk-Rec** advances the frontier of personalized AI by establishing a unified foundation for *interactive, explainable, and trustworthy recommendation*. It bridges the long-standing gap between the symbolic and the structural, between conversation and cognition, demonstrating that genuine personalization arises not from prediction alone but from *reasoning in dialogue*. By embedding large language models within graph-structured relational intelligence, the framework sets the stage for a new generation of **conversational reasoning systems**—systems that not only recommend but *understand*, not only predict but *explain*, and not only respond but *reason* alongside their users.

Future research inspired by this direction could give rise to **cognitive graph-LLM ecosystems** capable of collective reasoning across communities, domains, and modalities—moving recommendation science toward a future where human preference, machine intelligence, and relational understanding coevolve in transparent harmony.

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